Stage-conditional shift normalization for subject-robust EEG sleep staging

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Abstract

Automatic sleep staging models can lose accuracy when applied to unseen patients, yet a single label-free remedy is usually applied: replace the normalization statistics of the target recording. This assumes that subject shift is a recording-wide offset. We introduce a patient-identity diagnostic that tests this assumption by refitting a linear subject probe within each sleep stage. After marginal mean correction, the within-stage probe remains well above chance, whereas the overall probe is dominated by the removable offset. We therefore propose stage-conditional shift normalization (SCSN), a closed-form, label-free correction that estimates a separate shift for each stage from the model's soft posterior and subtracts the posterior-weighted shift during inference. SCSN contains marginal correction as its $\lambda = 0$ endpoint and applies unchanged to batch- and group-normalized networks. Across Sleep-EDF (39 recordings), HMC (151 recordings), two backbones and two normalization schemes, SCSN reduced the within-stage subject probe from 0.74 to 0.47 and from 0.53 to 0.21 on EEGNet. Classification accuracy was preserved on Sleep-EDF and improved by 1.4–1.9 points on HMC. On group-normalized models, SCSN significantly improved accuracy in three of four settings and improved worst-patient conformal coverage in all four. Cross-cohort transfer also exposes the method's precondition: a weak target posterior reduces SCSN to marginal correction, whereas oracle stages recover most of the gap.

Keywords: Sleep staging, Test-time adaptation, normalization, Subject variability, Electroencephalography, Conformal prediction

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1. Introduction

A deep network trained to stage sleep agrees with a human scorer about as well as two human scorers agree with each other, provided the patient comes from the cohort it was trained on. Applied to a patient from another clinic, the same network loses a substantial part of that agreement. The loss is not a failure of the filters: a slow oscillation is a slow oscillation in every laboratory. It is a failure of the statistics with which the network represents the patient, and those statistics are produced, to a large extent, by the normalization layers.

The default label-free remedy is marginal adaptation. Re-estimating the batch-normalization statistics on the recording being scored — adaptive batch normalization, AdaBN [1] — removes one mean and one variance per layer and per night, uses no label and changes no weight. It is cheap, deterministic and widely used. Its premise is that the patient is a recording-wide offset. This paper shows that the premise is only half true, and that the half it misses is the one that a single vector per recording cannot address.

Two parts of the patient. The class-conditional mean of the representation in a recording decomposes exactly into three terms (Section 3): the stage prototype μ_k the classifier was trained on, a recording-wide offset $a_{\mathcal{R}}$ that is the same for every epoch of the night, and a stage-conditional signature $s_{\mathcal{R},k}$ that describes how this patient's wake, N2 or REM differs from the prototype. The post-hoc mean subtraction that mimics AdaBN at the representation removes $a_{\mathcal{R}}$ and leaves $s_{\mathcal{R},k}$ untouched; layerwise AdaBN removes $a_{\mathcal{R}}$ and, as a nonlinear side effect, in some settings part of $s_{\mathcal{R},k}$, but it has no stage index and cannot target it. The residual is not small either way. Within each stage a linear probe still identifies the patient far above chance after layerwise AdaBN, and the conditional part of the residue is what the classifier compares when it decides between two stages: the logit margin under a marginal correction is perturbed by $s_{\mathcal{R},j} - s_{\mathcal{R},k}$.

Stage-conditional shift normalization. If the shift is indexed by the stage, it can be removed stage by stage. SCSN estimates, for each stage k and recording $\mathcal{R}$, the shift $\hat{\delta}_{\mathcal{R},k}$ between the recording's stage-conditional mean and the cohort prototype, and subtracts the posterior-weighted average of those shifts from every epoch: $\tilde{z}_i = z_i - \sum_k p_{ik}\hat{\delta}_{\mathcal{R},k}$, where p_{ik} is the model's

own softmax on the uncorrected representation. The estimator is closed-form with one shrinkage constant, uses no label, changes no weight, and needs no additional forward pass: the representation and the posterior come from the same pass over the night. The offset cancels for any posterior that sums to one; the stage-conditional signature cancels when the posterior is informative. Under the population correction the class-conditional means of every recording are mapped back to μ_k, so the logit margin between any two stages is restored to the one the head was trained on. Setting the mixing parameter of the interpolated form to zero recovers the marginal correction, so the two are endpoints of one family rather than unrelated methods.

Why the distinction matters in practice. The distinction is not only analytical. A batch-normalized stager can be adapted by replacing its statistics, and the test-time adaptation literature has produced a family of increasingly elaborate ways to do so [1, 2, 3, 4, 5]. A group-normalized stager cannot use the standard AdaBN replacement: its normalization is computed within an epoch and therefore has no test-time batch statistic to re-estimate. A bespoke entropy objective on GroupNorm's affine parameters could be constructed, but this is not the usual Tent procedure and we treat it as a separate method outside this study. Yet group normalization is an increasingly common choice precisely because it removes dependence on batch composition, and the representation it produces still carries the patient. On these checkpoints we compare the uncorrected model, the post-hoc marginal correction, and SCSN; SCSN is the only stage-aware method among the statistics-replacement approaches evaluated here.

The same failure separates SCSN from existing transfer methods for sleep and EEG. Cross-domain sleep staging has been addressed by test-time adaptation [6], whereas recent cross-subject models use meta-learning, domain alignment, or self-supervised representation learning [7, 8, 9]. These approaches improve subject generalization by changing training objectives or learned representations. SCSN leaves both the trained filters and the classifier unchanged and corrects only recording-level statistics that remain after training.

Measuring the right quantity. A subject-identification probe fitted to the penultimate representation is the natural diagnostic, but the overall probe is dominated by the marginal component: any per-recording centering collapses it, whether or not the conditional signature survives. We therefore refit

the probe *within each stage*. The within-stage probe is unchanged by a correction that removes only $a_{\mathcal{R}}$, and falls only when the correction removes $s_{\mathcal{R},k}$. The distinction is what separates the two families empirically, and it is the measurement we report throughout.

What kind of contribution this is. The contribution is analytical before it is algorithmic. The paper adds no layer, no loss and no training procedure; it characterizes what a trained stager's representation contains and which inference-time operations can change which part of it, and the algorithm follows from that characterization. Two results carry it. The first is an identity: the class-conditional means of a recording decompose exactly into a cohort prototype, a recording-wide offset and a stage-conditional signature, and the two parts are identifiable from each other by construction. The second is an impossibility statement, which we prove rather than observe: every correction that subtracts one vector per recording — an oracle that knows the patient's stage mix, an estimate of the recording mean, any vector learned from the recording — leaves the relative configuration of the stage-conditional means unchanged (Corollary 1). No better estimate of the marginal statistic can remove the signature, because the signature is not a statistic of the recording's mean; only an operation indexed by the stage can reach it. SCSN is then derived rather than designed: given the posterior, the posterior-weighted average of the per-stage shifts is the minimum-mean-square-error estimate of the shift that affects an epoch, and the residual is the posterior mass placed on the wrong stage (Remark 1). That it has a closed form is a consequence of the setting the method is meant for — a checkpoint whose weights are frozen and a night that carries no labels — and the closed form is the reason the method is deployable, not the reason it is new. Read as a normalization operator, SCSN is the class-conditional counterpart of AdaBN: its sufficient statistics are indexed by the classes the model assigns, rather than by the batch or by the epoch.

What the paper contributes.

1. A decomposition of the penultimate representation into a stage prototype, a recording offset and a stage-conditional subject signature, with propositions that state exactly what marginal adaptation leaves behind (Proposition 1) and what a stage-conditional correction removes (Proposition 2), and a corollary that no correction subtracting

one vector per recording can alter the relative geometry of the stage-conditional means (Corollary 1).

2. SCSN, a closed-form, label-free normalization of the representation that needs no retraining and no optimizer, applies unchanged to batch-normalized and group-normalized stagers, and contains the marginal correction as the endpoint $\lambda = 0$ of a one-parameter family (Algorithm 1).
3. A measurement protocol that separates the two components of patient identity, and evidence on two cohorts, two backbones and two normalization regimes that layerwise AdaBN leaves a large within-stage signature while SCSN removes several times more of it.
4. Evidence that on group-normalized stagers, where the standard AdaBN statistics replacement is undefined, SCSN remains applicable and removes the conditional signature; it also improves accuracy and per-patient conformal-coverage dispersion over the uncorrected model.
5. Controls that isolate the mechanism — mismatched statistics, orthogonalised corrections, hard pseudo-labels, low-rank restrictions — a Tent baseline evaluated on the same runs, and a cross-cohort experiment that delimits the regime in which a stage-conditional correction is useful.

The method is offline in the same sense as AdaBN: it reads the unlabeled epochs of the night it is about to score. Clinical staging is performed after the night, so this is the setting in which a deployed stager operates. No label, no epoch of another patient and no weight is used.

The paper reports where the method is not the best available one. On Sleep-EDF SleepCNN, layerwise AdaBN is 1.4 points more accurate than SCSN, a difference that is not significant; on Sleep-EDF EEGNet the two are indistinguishable; and Tent is competitive on the same cohort. On both HMC backbones SCSN has the highest mean accuracy of the arms, by a margin that is significant over the uncorrected model in both cases and not significant over layerwise AdaBN on HMC SleepCNN. The claim defended here is not that SCSN dominates every baseline on accuracy, but that it removes a component of patient identity that none of them removes, and that this component can be measured, predicted and corrected without labels.

Organisation. Section 2 places the work among test-time normalization and subject-variability methods. Section 3 derives the decomposition, states the propositions and defines the algorithm. Section 4 describes the cohorts, arms,

metrics and statistics. Section 5 reports the main comparison, the mechanism analyses, the ablations, the Tent baseline, the group-normalized setting and cross-cohort transfer. Section 6 discusses what the results do and do not license.

2. Related work

2.1. Normalization as a statistical interface

Batch normalization [10] is usually discussed for what it does during training. At inference it is a different object: a fixed affine map whose slope and offset were estimated on the training cohort, so that the representation of a test recording carries a term determined by how far that recording's statistics lie from the cohort's. Group normalization [11], instance normalization [12] and layer normalization [13] remove the batch from the computation and make the output of the layer depend only on the example it is applied to. Frankle et al. [14] show how much of a trained network's function is reproduced by its normalization parameters alone, and Nado et al. [15] evaluate prediction-time batch normalization under covariate shift; both locate a large part of the model's behavior in statistics rather than in filters.

Re-estimating the statistics on the test domain is adaptive batch normalization [1]. In sleep staging the operation is attractive because a night is a natural adaptation domain: its epochs are exchangeable enough to estimate a mean and variance, and no label is needed. What the operation cannot do is distinguish one sleep stage from another. It estimates one mean per recording, so any patient-dependent component that differs between stages survives it. This is the gap the present paper addresses.

A related line conditions normalization on a label. Conditional batch normalization and feature-wise transformations modulate the activations of a network with a class or text embedding [16, 17, 18]; the conditioning signal is known during training and supplied at inference. SCSN conditions on the stage as well, but it does not learn a conditioning map and it does not receive the stage: the stage is unknown at test time and is replaced by the model's own soft posterior. The weights stay frozen, and the only quantity estimated from the test recording is the shift itself.

The ingredients of the method are ordinary — estimate a mean, subtract it, index the estimate by a class — so the question is what the paper contributes beyond their assembly. The answer is the result that links

them. The decomposition of Section 3 proves that no correction subtracting one vector per recording can change the relative configuration of the stage-conditional means, however well that vector is estimated (Corollary 1); feature-wise modulation, in turn, is supplied with a class taken from an annotation, whereas in staging the class is unknown and the index has to be estimated from the model. The two observations only become one contribution when the first shows that the marginal route cannot work and the second shows what has to replace it. SCSN is the normalizer that results: class-conditional statistics, conditioned on a test-time estimate of the class rather than on a label. The experiments test the derivation rather than a combination — the interpolation between the marginal and conditional endpoints, the matched-versus-mismatched control, and the cross-cohort test in which a weak posterior should and does reduce the method to its marginal special case.

2.2. Test-time adaptation beyond marginal statistics

Tent [2] updates the affine parameters of normalization layers by minimizing the entropy of the posterior on the test recording; test-time training [19] uses a self-supervised objective. Both change weights with an optimizer. CAFA [3] aligns target features to the source distribution with a class-aware objective, and UnMix-TNS [4] improves the estimation of normalization statistics when test labels are temporally correlated. PSDNorm [5] replaces mean-and-variance normalization by a test-time alignment of the power spectral density of each feature map and reports strong cross-dataset sleep staging; the operation is input-domain and frequency-resolved, and it does not use the predicted stage. FairAdaBN [20] makes adaptive normalization attribute-aware for fairness.

SCSN differs from this family in three respects. It is closed-form: no optimizer, step size or number of steps is chosen at test time. It is stage-conditional: the statistic it removes is indexed by the sleep stage, and the index comes from the model's own posterior rather than from a label or a frequency band. And its target is explicit: the class-conditional mean of the representation, which is the quantity the classifier's decision boundaries were fitted around. The interpolated form in Section 3 contains the post-hoc marginal correction as the endpoint $\lambda = 0$, so the comparison is a nested one rather than a comparison of unrelated methods. Cross-domain sleep staging has also been addressed directly by test-time adaptation [6]. The distinction from SCSN is the conditioning variable: these methods adapt a shared

target-domain or test-domain transformation, whereas SCSN updates a separate first-moment correction for each posterior-indexed sleep stage before the shared classifier is applied.

The family also has a blind spot that matters for the present paper. AdaBN, Tent and their descendants operate on batch-normalization statistics or affine normalization parameters. A group-normalized stager has no batch statistic to replace; a bespoke entropy objective on its affine parameters would be a different method from standard Tent and is not evaluated here. Group normalization is an increasingly common choice precisely because it removes dependence on batch composition, so the lack of a comparable statistics-replacement correction is a practical gap. SCSN is defined on the representation and therefore applies unchanged. The other correction compared for such a checkpoint is post-hoc centering, which removes one vector per recording; the experiments show that this removes the overall subject probe while moving the within-stage probe by no more than a third of what SCSN removes.

2.3. Subject variability in automatic sleep staging

Deep stagers have moved from single-cohort evaluation [21, 22] to multi-cohort and cross-cohort settings [23, 24], and the gap between within-cohort and cross-cohort accuracy has become an object of study. Nasiri and Clifford [25] show that population subgrouping improves staging for under-represented patients. Adversarial subject classifiers [26] and disentangled architectures [27] change the weights so that the representation carries less of the subject; both require retraining and, in the disentangled case, a subject-invariance objective. Kim et al. [28] use subject-aware batch normalization during training for cross-subject brain decoding. Normalization has also been used as a cross-subject alignment device for EEG [29], usually on the raw signal rather than inside the network. Recent Neurocomputing studies target subject generalization through few-shot meta-learning [7], semi-supervised domain adaptation [8], and generalizable representation learning for EEG [9]. Like the methods above, these approaches alter training or the learned representation; the inference-time decomposition studied here is complementary and can be applied to their frozen outputs.

These methods answer the question *how do we train a model that does not encode the subject*. The present paper answers a different question: given a model that already encodes the subject, which part of the encoding is removable at inference, and which part is not. The answer separates a

recording-wide offset from a stage-conditional signature; the first is removed by marginal adaptation and the second by SCSN, and only the second requires a stage posterior. A disentangled architecture can be combined with SCSN, and we test SCSN on group-normalized checkpoints for exactly that reason.

2.4. Uncertainty and patient-level reliability

Sleep staging is scored with a set of stages, and disagreement on boundary epochs is documented [30]. Calibration measures whether a predicted probability means what it says [31]; conformal prediction produces a set with a finite-sample coverage guarantee [32, 33]. Adaptive and regularized scores tighten the sets [34, 35], and the guarantee has been extended to time series and distribution drift [36, 37].

The guarantee is marginal over the exchangeable unit. When the unit is the patient, a stager can honour it on average while under-covering particular patients, and a representation that separates patients concentrates the failures. We therefore report per-patient coverage, its dispersion and the worst-covered patient, and we ask whether the correction that removes the patient signature also equalises the coverage. The question is not whether the guarantee is restored — it remains marginal — but whether the mechanism that concentrates the failures is the mechanism that carries the patient.

3. Stage-conditional shift normalization

3.1. What the normalization layer leaves in the representation

Let $z_i \in \mathbb{R}^d$ be the penultimate representation of epoch i, let $k_i \in \{1, \dots, K\}$ be its sleep stage, and let $r(i)$ be the recording it belongs to. Write $\mathcal{R}$ for a recording and $\pi_{\mathcal{R},k}$ for the fraction of its epochs scored as stage k. The class-conditional mean of the representation in a recording is

$$m_{\mathcal{R},k} \;=\; \mathbb{E}[z_i \mid r(i) = \mathcal{R},\, k_i = k]\,, \qquad m_{\mathcal{R}} \;=\; \sum_k \pi_{\mathcal{R},k}\, m_{\mathcal{R},k}, \tag{1}$$

and the cohort mean is $\mu_{\text{coh}} = \sum_k \pi_k \mu_k$, with μ_k the class mean over the training cohort and π_k its stage mix.

The decomposition that organises the paper is a definition, not an assumption. Set

$$a_{\mathcal{R}} \;=\; m_{\mathcal{R}} - \bar{\mu}_{\mathcal{R}}, \qquad s_{\mathcal{R},k} \;=\; m_{\mathcal{R},k} - \mu_k - a_{\mathcal{R}}, \qquad \bar{\mu}_{\mathcal{R}} = \sum_k \pi_{\mathcal{R},k}\mu_k. \quad (2)$$

Then $m_{\mathcal{R},k} = \mu_k + a_{\mathcal{R}} + s_{\mathcal{R},k}$ and, by construction, $\sum_k \pi_{\mathcal{R},k} s_{\mathcal{R},k} = 0$. The three terms have different meanings. μ_k is the stage prototype the classifier was trained on. $a_{\mathcal{R}}$ is a recording-wide offset: it is the same vector for every epoch of the night and is what a batch-normalization layer manufactures when the recording's first and second moments differ from the cohort's. $s_{\mathcal{R},k}$ is the part of the patient that is not an offset: within a stage, this patient's spindles, slow waves or alpha differ from the cohort prototype in a way that is stable across the night. The constraint makes the offset and the conditional signature identifiable from each other.

3.2. Marginal adaptation removes the offset and keeps the signature

The default label-free remedy is to replace the cohort statistics by the recording's, either by re-estimating every batch-normalization statistic (AdaBN) or, at the representation, by subtracting the recording's mean shift $m_{\mathcal{R}} - \mu_{\text{coh}}$. The next proposition states exactly what that leaves behind.

Proposition 1 (Marginal residual)**.** *Under the decomposition* (2)*, marginal correction* $z_i \mapsto z_i - (m_{\mathcal{R}} - \mu_{\text{coh}})$ *gives*

$$\mathbb{E}[z_i - (m_{\mathcal{R}} - \mu_{\text{coh}}) \mid r(i) = \mathcal{R}, k_i = k] \;=\; \mu_k + s_{\mathcal{R},k} + \big(\mu_{\text{coh}} - \bar{\mu}_{\mathcal{R}}\big). \quad (3)$$

Proof. Substitute $m_{\mathcal{R}} = \bar{\mu}_{\mathcal{R}} + a_{\mathcal{R}}$ from (2) into the conditional mean $m_{\mathcal{R},k} = \mu_k + a_{\mathcal{R}} + s_{\mathcal{R},k}$ and cancel $a_{\mathcal{R}}$. □

Two consequences follow. The conditional signature $s_{\mathcal{R},k}$ survives untouched: marginal adaptation removes a vector that is constant across stages and cannot remove one that is not. And the correction adds $\mu_{\text{coh}} - \bar{\mu}_{\mathcal{R}}$, which is zero only when the recording's stage mix equals the cohort's. A recording with an atypical architecture therefore keeps a residual offset that is a function of *how it slept*, not of who the patient is. Even an oracle marginal correction that knows $\bar{\mu}_{\mathcal{R}}$ and subtracts $m_{\mathcal{R}} - \bar{\mu}_{\mathcal{R}}$ would leave $\mu_k + s_{\mathcal{R},k}$; the conditional signature is out of reach of any operation that uses one mean per recording. The next corollary states that this is not a property of a particular estimate of the mean but of the entire family.

Corollary 1 (No single-vector correction removes the signature)**.** *Let g be any function of the recording alone, and apply the correction $z_i \mapsto z_i - g(\mathcal{R})$. Then for any two stages j, k,*

$$\begin{aligned}&\mathbb{E}[z_i - g(\mathcal{R}) \mid r(i) = \mathcal{R}, k_i = j] \\ &\quad - \mathbb{E}[z_i - g(\mathcal{R}) \mid r(i) = \mathcal{R}, k_i = k] = (\mu_j - \mu_k) + (s_{\mathcal{R},j} - s_{\mathcal{R},k}),\end{aligned} \tag{4}$$

independently of g.

Proof. The same vector $g(\mathcal{R})$ is subtracted from both conditional means, and $m_{\mathcal{R},j} - m_{\mathcal{R},k} = (\mu_j - \mu_k) + (s_{\mathcal{R},j} - s_{\mathcal{R},k})$ by (2). □

The corollary is the precise sense in which the signature is out of reach of the marginal family. Whatever g is — the recording mean, an oracle that knows the patient's stage mix, or a vector learned from the recording — the displacement the head compares between two stages changes by the same $s_{\mathcal{R},j} - s_{\mathcal{R},k}$. A better estimate of the marginal statistic moves the offset and leaves the relative geometry exactly where it was; only an operation whose index depends on k can change (4). The statement is a consequence of the definition rather than of a modeling assumption: the conditional means are what they are, and the difference between two of them contains $s_{\mathcal{R},j} - s_{\mathcal{R},k}$ exactly. What the corollary rules out is therefore not a particular estimator but a class of interventions.

The consequence for the classifier is equally direct. The logit margin between two stages under marginal correction is

$$W\big(m'_{\mathcal{R},j} - m'_{\mathcal{R},k}\big) = W(\mu_j - \mu_k + s_{\mathcal{R},j} - s_{\mathcal{R},k})\,, \tag{5}$$

so the patient's stage-conditional signature is added to the very quantity the classifier compares. When that signature is large — an atypical patient, a recording with unusual stage-specific morphology — the margin can move across the decision boundary, and accuracy falls. This is the mechanism behind the accuracy difference between the marginal and conditional corrections in Section 5.2.

3.3. Stage-conditional shift normalization

The alternative is to estimate the shift separately for each stage and to remove it with the model's own soft assignments. Let $p_{ik} = \Pr(k_i = k \mid z_i)$

be the posterior of the unadapted model, and let

$$\hat{m}_{\mathcal{R},k} = \frac{\sum_{i:r(i)=\mathcal{R}} p_{ik} z_i + \lambda_s \mu_k}{\sum_{i:r(i)=\mathcal{R}} p_{ik} + \lambda_s}, \qquad \hat{\delta}_{\mathcal{R},k} = \hat{m}_{\mathcal{R},k} - \mu_k, \tag{6}$$

where $\lambda_s \geq 0$ is a pseudo-count that shrinks rare stages towards the cohort prototype; in all experiments $\lambda_s = 10$. The corrected representation is

$$\tilde{z}_i \;=\; z_i - \sum_k p_{ik}\, \hat{\delta}_{\mathcal{R},k}. \tag{7}$$

Equations (6)–(7) are stage-conditional shift normalization (SCSN). The operation is a closed-form post-processing of one forward pass: the pass gives p_{ik} and z_i, the per-stage shifts are accumulated, and the corrected representation is obtained by subtracting the weighted shifts. No second pass over the network is required. No weight is updated, no optimizer is run, no label of the recording is read, and no subject identifier is used.

Read as a normalization operator, (7) has the same shape as the normalizers reviewed in Section 2 but different sufficient statistics: batch normalization uses the moments of a batch, group normalization the moments of an epoch, and SCSN the class-conditional means of the recording, weighted by the model's posterior. It is a normalization whose statistics are indexed by the classes the model assigns.

For a recording of n epochs and penultimate dimension d, accumulating the stage summaries takes $O(nKd)$ arithmetic and $O(Kd)$ additional storage; the correction has the same arithmetic complexity. No network replay, backward pass, or gradient is required.

Proposition 2 (Conditional residual)**.** *Let $\hat{\delta}_{\mathcal{R},k} = \delta_{\mathcal{R},k} = a_{\mathcal{R}} + s_{\mathcal{R},k}$ (the population value). Then for every epoch,*

$$\tilde{z}_i = z_i - \sum_k p_{ik}\big(a_{\mathcal{R}} + s_{\mathcal{R},k}\big) = z_i - a_{\mathcal{R}} - \sum_k p_{ik} s_{\mathcal{R},k}. \tag{8}$$

In particular the offset cancels for any *posterior with $\sum_k p_{ik} = 1$, and if $p_{ik} = \mathbb{1}[k_i = k]$ the conditional mean of $\tilde{z}_i$ is exactly μ_k.*

Proof. $\sum_k p_{ik} = 1$ gives $\sum_k p_{ik} a_{\mathcal{R}} = a_{\mathcal{R}}$. For hard assignments $\sum_k p_{ik} s_{\mathcal{R},k} = s_{\mathcal{R},k_i}$, and $\mu_k + a_{\mathcal{R}} + s_{\mathcal{R},k} - a_{\mathcal{R}} - s_{\mathcal{R},k} = \mu_k$. □

For a soft posterior the residual is

$$\mathbb{E}[\tilde{z}_i \mid r(i) = \mathcal{R}, k_i = k] = \mu_k + \sum_j \big(\mathbb{1}[j = k] - p_{ij}\big) s_{\mathcal{R},j}, \tag{9}$$

so the error is the part of the conditional signature assigned to the wrong stage. It vanishes when the posterior is correct and is bounded by $\max_j \|s_{\mathcal{R},j}\| \sum_j |\mathbb{1}[j = k] - p_{ij}|$. This is why the method is robust to an imperfect but informative posterior and why it degrades when the posterior is close to uniform, a prediction we test in the cross-cohort experiment.

Remark 1 (Why the posterior-weighted average). Condition on the recording and write $\Delta_i = a_{\mathcal{R}} + s_{\mathcal{R},k_i}$ for the shift that affects epoch i. Given z_i, the least-mean-square estimate of Δ_i among functions of z_i is the conditional expectation

$$\mathbb{E}[\Delta_i \mid z_i] = a_{\mathcal{R}} + \sum_k \Pr(k_i = k \mid z_i)\, s_{\mathcal{R},k}, \tag{10}$$

and (7) subtracts this quantity with the model's posterior in place of the unknown true one. The correction is therefore not a heuristic blend: it is the estimate that minimizes the expected squared residual, and the residual it leaves, (9), is the part of the shift attributed to the wrong stage. The hard-assignment variant replaces the posterior by a point mass; it has a smaller conditional bias and a larger variance, and the experiments show that it over-corrects. The interpolated family $z_i - \delta_{\mathcal{R}} - \lambda(\sum_k p_{ik}\hat{\delta}_{\mathcal{R},k} - \delta_{\mathcal{R}})$ traces a path from ignoring the stage ($\lambda = 0$) to trusting the posterior completely ($\lambda = 1$).

The class geometry is preserved by the same argument. Under the population correction the class-conditional means of the corrected representation are μ_k, independently of the recording, so

$$W(\tilde{m}_{\mathcal{R},j} - \tilde{m}_{\mathcal{R},k}) = W(\mu_j - \mu_k) \tag{11}$$

for every recording: the logit margin the classifier was trained on is restored, whereas under marginal correction it is perturbed by $s_{\mathcal{R},j} - s_{\mathcal{R},k}$ as in (5). This is the formal content of the accuracy result: SCSN does not merely remove a nuisance, it maps every recording back to the class geometry on which the head was fitted.

Two controls follow from the algebra. If the correction is constrained to the null space of W, the logits are unchanged exactly, so any change in identity cannot be attributed to a change in predictions; we implement this

Algorithm 1 Stage-conditional shift normalization (one recording)

Require: frozen stager $f = W\phi + b$, unlabeled epochs $x_1, \ldots, x_n$ of a recording $\mathcal{R}$

Require: cohort class means μ_k and cohort mean μ_{coh}

$z_i \leftarrow \phi(x_i)$ ▷ forward pass

$p_{ik} \leftarrow \text{softmax}(W z_i + b)$

for $k = 1, \ldots, K$ **do**

  $\hat{m}_{\mathcal{R},k} \leftarrow (\sum_i p_{ik} z_i + \lambda_s \mu_k)/(\sum_i p_{ik} + \lambda_s)$

  $\hat{\delta}_{\mathcal{R},k} \leftarrow \hat{m}_{\mathcal{R},k} - \mu_k$

end for

$\tilde{z}_i \leftarrow z_i - \sum_k p_{ik} \hat{\delta}_{\mathcal{R},k}$ ▷ correction

return $\text{softmax}(W \tilde{z}_i + b)$

by projecting each $\hat{\delta}_{\mathcal{R},k}$ off the row space of W. If the shifts of a *different* recording are applied, the correction is a non-zero recording-specific vector and the identity is replaced rather than removed; this is the matched-versus-mismatched control.

Remark 2 (Estimation error). With $n_{\mathcal{R},k} = \sum_i p_{ik}$ effective epochs, the shrunk estimator (6) has bias at most $\lambda_s \|m_{\mathcal{R},k} - \mu_k\|/(n_{\mathcal{R},k} + \lambda_s)$ towards the cohort prototype and standard deviation of order $\|s_{\mathcal{R},k}\|/\sqrt{n_{\mathcal{R},k}}$. Both terms are small for the stages that dominate a night (N2, REM) and large for N1, which is why the shrinkage is kept in all experiments. On HMC a recording averages about 900 epochs, of which roughly 100 are scored N1 and 330 N2, so a pseudo-count of ten moves the N1 estimate by at most a tenth towards the prototype and the N2 estimate by at most a thirtieth. The estimator is a soft, regularized version of the stage-conditional sample mean; no gradient, iteration or held-out label enters it.

3.4. Relation to existing normalizers

Batch normalization and AdaBN. A batch-normalized stager at inference is an affine map with cohort statistics. AdaBN replaces those statistics by the recording's at every layer [1]. That is a nonlinear, layer-wise operation and is not the same object as the representation-level marginal correction of Proposition 1; the proposition describes the post-hoc operation that subtracts one vector from the frozen penultimate representation, which is the operation SCSN generalises. We report both: the post-hoc marginal arm isolates the

algebra, and layerwise AdaBN is the stronger inference-time baseline that a practitioner with access to the model would use. Setting $\lambda = 0$ in the interpolated form $z_i - \delta_{\mathcal{R}} - \lambda(\sum_k p_{ik}\hat{\delta}_{\mathcal{R},k} - \delta_{\mathcal{R}})$ recovers the post-hoc marginal correction; $\lambda = 1$ is SCSN. We report the whole λ path.

Group and instance normalization. Group normalization removes the batch from the computation [11]: the output of a group-normalized layer depends only on the epoch it is applied to. Standard AdaBN therefore has no batch statistic to replace. Standard Tent updates batch-normalization affine parameters; an entropy objective on the affine parameters of group normalization would be a different test-time method and is not evaluated here. It does not follow that the learned representation is free of the recording. Group normalization removes the per-epoch mean at each layer, but the network that follows maps the remaining recording-dependent variation into the penultimate space; the experiments below show that a linear probe still identifies the patient from a group-normalized representation, and that the post-hoc marginal correction still changes both accuracy and calibration. The post-hoc marginal correction can remove only the recording-wide offset, and the experiments show that it moves the within-stage probe by no more than a third of what SCSN removes. SCSN is the only stage-aware statistics-replacement method evaluated there, and we therefore apply it to these checkpoints.

Class-conditional and entropy-based test-time adaptation. Tent minimizes the entropy of the posterior over the affine parameters of the normalization layers [2]; it is class-agnostic and optimizes a surrogate. SCSN has a closed form, one hyperparameter (the shrinkage), and a target: the per-stage mean. The two are compared directly in Section 5.4.

Subject-invariant architectures. Adversarial subject classifiers and subject identifiers [26, 24] change the weights and need retraining; disentangled architectures [23] do the same. SCSN is applied to a frozen checkpoint and changes no parameter. It is therefore not a competitor to those methods but a post-hoc alternative for models that already exist, and it can be stacked on top of any of them.

4. Experimental protocol

4.1. Cohorts

Sleep-EDF. The first cohort is the Sleep-EDF expanded database [38]: 20 healthy adults, 39 overnight polysomnograms, one EEG derivation (Fpz-Cz)

recorded at 100 Hz and scored in 30 s epochs under the R&K rules [39]. Wake, N1, N2, N3 and REM are retained; movement and unscoreable epochs are dropped. Signals are band-pass filtered between 0.5 and 30 Hz, resampled to 100 Hz and standardised per recording, so that the raw mean and variance of a night carry no information about the patient into the network.

HMC. The second cohort is the Haaglanden Medisch Centrum sleep staging database [40]: 151 overnight recordings from 151 patients, one EEG channel (C4-M1) recorded at 256 Hz and scored in 30 s epochs under the AASM rules. The same five classes are retained, the signal is band-pass filtered between 0.5 and 30 Hz, resampled to 100 Hz and standardised per recording. The processed cohort contains 137,243 labeled epochs, between 470 and 1,311 per recording. It differs from the first cohort in population, montage, equipment, raw sampling rate and scoring manual, which is what makes it a test of whether a statement about patients survives a change of cohort.

4.2. Backbones

Two convolutional backbones are used. EEGNet [41], adapted to a single channel, has 1,445 parameters; SleepCNN is a four-block VGG-style one-dimensional network with 256 channels in its last block and 2.24×10^5 parameters. Both were trained with batch normalization; a group-normalized copy of each was trained separately for the compatibility experiment. Architectures, optimization and splits are identical between the batch and group copies except for the normalization layer.

4.3. Arms

All arms are evaluated on the same checkpoint, the same split and the same test epochs; only the inference-time transformation of the penultimate representation differs.

BN. The trained checkpoint with its cohort statistics, no correction.

Marginal. The post-hoc representation-level correction $z - \widehat{\delta}_{\mathcal{R}}$, with $\widehat{\delta}_{\mathcal{R}}$ the difference between the recording's mean representation and the training cohort's. It is the representation-level counterpart of AdaBN and the operation SCSN generalises; unlike AdaBN it applies to a group-normalized checkpoint, because it reads only the penultimate features.

AdaBN. The same checkpoint with the statistics of every normalization layer replaced by those of the recording being scored [1]. This is a nonlinear, layer-wise operation; the representation-level marginal correction of Proposition 1 is the post-hoc arm *Marginal* below, which removes one vector per recording and is the operation SCSN generalises.

SCSN. Stage-conditional shift normalization of Equations (6)–(7), $\lambda = 1$, $\lambda_s = 10$. The posterior of the uncorrected model is used.

AdaBN+SCSN. The stacked arm: layerwise AdaBN first, then SCSN in the adapted feature space, with prototypes estimated on source subjects that have each been adapted to their own recording in the same way. It tests whether the two corrections remove the same component or different ones.

SCSN-λ. The interpolated correction $z - \delta_{\mathcal{R}} - \lambda(\sum_k p_k \hat{\delta}_{\mathcal{R},k} - \delta_{\mathcal{R}})$ for $\lambda \in \{0.25, 0.5, 0.75\}$; $\lambda = 0$ is the post-hoc marginal correction and $\lambda = 1$ is SCSN.

SCSN-r. The correction restricted to the top-r principal directions of the between-subject, within-stage covariance estimated on the training subjects, for $r \in \{1, 2, 4\}$.

SCSN-hard. The same estimator with one-hot pseudo-labels.

SCSN-orth. The correction projected off the row space of the classifier, so that the logits are unchanged exactly.

Mismatched. The shifts of another recording of the same fold, applied cyclically; the control for whether the effect requires the recording's own statistics.

Tent. Entropy minimization over the affine parameters of the normalization layers [2], with the published optimizer (SGD, momentum 0.9, learning rate 2.5×10^{-4}, one pass over the recording), and two Adam variants (one and five steps, learning rate 10^{-3}). Calibration recordings are adapted in the same way as test recordings.

The group-normalized checkpoints are evaluated with the arms that do not require batch statistics: the uncorrected model, the post-hoc marginal

correction and SCSN. Standard AdaBN has no batch statistic to re-estimate on these checkpoints. Standard Tent updates batch-normalization affine parameters; a separate GroupNorm-specific entropy objective is not evaluated here because it would introduce a new optimizer, learning-rate choice, and step-count protocol rather than measuring the effect of stage conditioning.

4.4. Training and splits

Evaluation uses five subject-wise folds. Within a fold the 20% of subjects with the lowest hash are held out for testing, a further disjoint 20% form the conformal calibration set, 15% of the remaining subjects are used for early stopping, and the rest for training. No subject appears in more than one of the four groups of a fold. On Sleep-EDF each fold is run with three seeds for EEGNet and one seed for SleepCNN, giving fifteen and five runs per arm; on HMC each fold is run with two seeds for both backbones, giving ten runs per arm. Optimization uses Adam with learning rate 10^{-3}, weight decay 10^{-4} and cross-entropy without class weights, for at most 40 epochs with early stopping on the validation split (patience 8). The best validation checkpoint is kept.

4.5. Conformal evaluation

For each run, the calibration subjects' epochs fit a split-conformal predictor with the cumulative (APS) non-conformity score [34] at $\alpha = 0.1$, and the test subjects' epochs are predicted as sets. We report coverage, mean set size, and review workload (the fraction of epochs whose set holds more than one stage). Because the guarantee is marginal over patients, coverage is also computed per test subject, from which we report the worst-covered subject, the standard deviation of per-subject coverage, the 10th percentile and the median. The calibration split is disjoint from the test split, and the quantile is refitted after every correction.

4.6. Identity measurements

For every run the penultimate representation of every test epoch is extracted under each arm, and two probes are fitted. The *overall* probe is a multinomial logistic regression from the representation to the patient index, fitted with three-fold cross-validation over epochs. The *within-stage* probe repeats the fit separately for the epochs of each stage and averages the five accuracies over the stages that contain at least fifty epochs. The overall probe is sensitive to any recording-specific offset, including the one that a

marginal correction adds; the within-stage probe holds the stage fixed, so it measures the part of the identity that is not a difference in sleep architecture and does not react to a correction that removes only a recording-wide mean. To keep the cost bounded on the large cohort, a fixed random subsample of at most 5,000 epochs per run (3,000 for the 256-dimensional SleepCNN features) is used; the subsample is drawn with a fixed seed, so every arm of a run is probed on exactly the same epochs. Chance is $1/|\mathcal{S}|$ for $|\mathcal{S}|$ held-out subjects: four on Sleep-EDF and thirty-one on HMC. Both probes use the test patients' identifiers, and the within-stage probe additionally uses the true stage of each test epoch to form the groups. They are diagnostics of the representation, not components of any arm: no correction reads a test label, and the probe is fitted after the arm has produced its representation.

4.7. Statistics

All comparisons between arms are paired: the arms of a run share the checkpoint, the fold, the seed, the test epochs and the probe subsample. Reported differences are the mean of the per-run differences, a 95% confidence interval from the t distribution over runs, the standardised mean difference d_z, and a two-sided Wilcoxon signed-rank test. Because the seeds of a fold share their test subjects, runs within a fold are not independent; every paired effect is therefore also reported as a sign count over folds (the sign of the fold-averaged difference). p-values are unadjusted, and the number of paired runs behind every contrast is reported so that the evidence can be weighed.

5. Results

All arms of a run share the checkpoint, the fold, the seed, the test epochs and the probe subsample, so every comparison below is paired. Reported differences are means of per-run differences with a 95% confidence interval and a two-sided Wilcoxon p-value; n is the number of paired runs. Tables 1 and 2 collect the arm means for the four settings of each backbone, and Table 3 collects the key paired contrasts. "Probe" is the subject-identification accuracy of a linear readout of the penultimate representation, "within" is the same probe refitted inside each stage and averaged over stages, and chance is 0.25 for the four held-out patients of a Sleep-EDF fold and $1/31 \approx 0.032$ for the thirty-one recordings of an HMC fold. "κ", "MF1" and "ECE" are Cohen's κ, the macro-averaged F_1 and the expected calibration error of the

Table 1: EEGNet: mean over runs per arm (five folds × three seeds on Sleep-EDF, five folds × two seeds on HMC). "Probe" and "Within" are subject identification overall and within stage; "κ" and "MF1" are Cohen's κ and the macro-averaged F_1; "ECE" is the expected calibration error; "Worst" and "SD" are the minimum and standard deviation of per-patient conformal coverage at $\alpha = 0.1$. Standard AdaBN and the batch-normalization Tent procedure are not applicable to group-normalized checkpoints and are omitted there.

Cohort	Arm	n	Acc	κ	MF1	ECE	Probe	Within	Worst	SD
Sleep-EDF BN	BN	15	0.781	0.698	0.644	0.030	0.715	0.738	0.828	0.055
	Marginal	15	0.777	0.686	0.642	0.033	0.301	0.735	0.838	0.047
	AdaBN	15	0.782	0.696	0.645	0.043	0.466	0.700	0.866	0.028
	SCSN	15	0.784	0.701	0.645	0.044	0.314	0.465	0.857	0.036
	SCSN-$\lambda_{0.5}$	15	0.788	0.707	0.648	0.034	0.529	0.598	0.845	0.043
	Tent-SGD	15	0.782	0.698	0.647	0.038	0.384	0.714	0.870	0.023
HMC BN	BN	10	0.672	0.564	0.594	0.028	0.385	0.527	0.649	0.086
	Marginal	10	0.648	0.522	0.571	0.035	0.074	0.411	0.599	0.096
	AdaBN	10	0.674	0.563	0.591	0.029	0.212	0.473	0.698	0.069
	SCSN	10	0.690	0.587	0.613	0.033	0.073	0.209	0.678	0.067
	SCSN-$\lambda_{0.5}$	10	0.689	0.586	0.610	0.034	0.205	0.340	0.680	0.072
	Tent-SGD	10	0.680	0.570	0.597	0.025	0.179	0.453	0.673	0.070
Sleep-EDF GN	BN	15	0.770	0.683	0.634	0.041	0.541	0.638	0.855	0.038
	Marginal	15	0.778	0.690	0.637	0.031	0.298	0.616	0.863	0.029
	AdaBN	–	–	–	–	–	–	–	–	–
	SCSN	15	0.780	0.696	0.640	0.044	0.330	0.449	0.864	0.034
	SCSN-$\lambda_{0.5}$	–	–	–	–	–	–	–	–	–
	Tent-SGD	–	–	–	–	–	–	–	–	–
HMC GN	BN	10	0.693	0.595	0.628	0.026	0.231	0.369	0.695	0.067
	Marginal	10	0.697	0.593	0.632	0.017	0.057	0.315	0.724	0.059
	AdaBN	–	–	–	–	–	–	–	–	–
	SCSN	10	0.709	0.616	0.643	0.017	0.065	0.156	0.708	0.059
	SCSN-$\lambda_{0.5}$	–	–	–	–	–	–	–	–	–
	Tent-SGD	–	–	–	–	–	–	–	–	–

posterior. "Worst" and "SD" are the minimum and the standard deviation of per-patient conformal coverage at $\alpha = 0.1$.

5.1. One patient, two measurable components

Under the cohort statistics, a linear readout of the penultimate representation identifies the patient with an accuracy of 0.715 on Sleep-EDF EEGNet and 0.385 on HMC EEGNet, against chance levels of 0.25 and 0.032 (Table 1). Refitted within each stage, the probe is 0.738 and 0.527: most of the identity is not an artifact of patients having different sleep architectures, because it survives holding the stage fixed.

Table 2: SleepCNN, same layout as Table 1; Sleep-EDF SleepCNN uses one seed per fold (five runs per arm) and HMC uses two (ten runs per arm).

Cohort	Arm	n	Acc	κ	MF1	ECE	Probe	Within	Worst	SD
Sleep-EDF BN	BN	5	0.803	0.730	0.709	0.038	0.847	0.848	0.840	0.048
	Marginal	5	0.805	0.729	0.696	0.024	0.298	0.865	0.870	0.028
	AdaBN	5	0.814	0.738	0.719	0.029	0.773	0.848	0.869	0.025
	SCSN	5	0.800	0.725	0.699	0.043	0.319	0.550	0.847	0.034
	SCSN-$\lambda_{0.5}$	5	0.805	0.732	0.707	0.037	0.634	0.694	0.845	0.040
	Tent-SGD	5	0.818	0.746	0.715	0.026	0.766	0.858	0.871	0.023
HMC BN	BN	10	0.698	0.598	0.641	0.034	0.681	0.805	0.645	0.075
	Marginal	10	0.663	0.541	0.602	0.068	0.126	0.720	0.711	0.061
	AdaBN	10	0.702	0.598	0.647	0.030	0.324	0.723	0.649	0.072
	SCSN	10	0.712	0.615	0.654	0.036	0.090	0.374	0.664	0.066
	SCSN-$\lambda_{0.5}$	–	–	–	–	–	–	–	–	–
	Tent-SGD	–	–	–	–	–	–	–	–	–
Sleep-EDF GN	BN	5	0.806	0.732	0.709	0.026	0.821	0.832	0.860	0.035
	Marginal	5	0.816	0.744	0.715	0.021	0.295	0.889	0.864	0.029
	AdaBN	–	–	–	–	–	–	–	–	–
	SCSN	5	0.817	0.747	0.718	0.035	0.323	0.583	0.870	0.025
	SCSN-$\lambda_{0.5}$	–	–	–	–	–	–	–	–	–
	Tent-SGD	–	–	–	–	–	–	–	–	–
HMC GN	BN	10	0.719	0.627	0.665	0.025	0.528	0.663	0.677	0.070
	Marginal	10	0.698	0.593	0.641	0.031	0.122	0.745	0.724	0.058
	AdaBN	–	–	–	–	–	–	–	–	–
	SCSN	10	0.728	0.639	0.674	0.025	0.088	0.341	0.697	0.059
	SCSN-$\lambda_{0.5}$	–	–	–	–	–	–	–	–	–
	Tent-SGD	–	–	–	–	–	–	–	–	–

The post-hoc marginal correction changes the two measurements in opposite ways. On Sleep-EDF it collapses the overall probe from 0.715 to 0.301 and leaves the within-stage probe at 0.735, a difference of -0.002 $[-0.029, +0.025]$, $p = 0.85$. On HMC it collapses the overall probe from 0.385 to 0.074 and reduces the within-stage probe only to 0.411. The overall probe is therefore not a sufficient statistic for the presence of the patient: a single vector subtraction per recording removes the direction that dominates it, whether or not the stage-conditional structure survives. The within-stage probe is the measurement that distinguishes the two corrections, and it is the one we use in the rest of the section.

Table 3: Paired differences for the key contrasts, with the Wilcoxon p-value in parentheses. Upper block: accuracy. Lower block: within-stage subject probe. A negative probe difference means less identity.

Contrast	edf bn eegnet	edf bn sleepcnn	hmc bn eegnet	hmc bn sleepcnn
Accuracy difference				
SCSN − BN	+0.002 (0.359)	-0.004 (0.812)	+0.019 (0.00195)	+0.014 (0.00391)
SCSN − Marginal	+0.007 (0.252)	-0.005 (0.812)	+0.042 (0.00195)	+0.049 (0.00195)
SCSN − AdaBN	+0.002 (0.89)	-0.014 (0.812)	+0.016 (0.00195)	+0.010 (0.084)
AdaBN − BN	+0.001 (0.804)	+0.010 (0.625)	+0.002 (0.846)	+0.004 (0.695)
Marginal − BN	-0.004 (0.561)	+0.002 (1)	-0.023 (0.00977)	-0.035 (0.00391)
Within-stage probe difference				
SCSN − BN	-0.273 (6.1e-05)	-0.298 (0.0625)	-0.318 (0.00195)	-0.431 (0.00195)
SCSN − Marginal	-0.270 (6.1e-05)	-0.316 (0.0625)	-0.202 (0.00195)	-0.346 (0.00195)
SCSN − AdaBN	-0.235 (6.1e-05)	-0.298 (0.0625)	-0.264 (0.00195)	-0.349 (0.00195)
AdaBN − BN	-0.038 (0.0103)	+0.000 (1)	-0.054 (0.00195)	-0.083 (0.00195)
Marginal − BN	-0.002 (0.847)	+0.018 (0.125)	-0.116 (0.00195)	-0.085 (0.00195)

5.2. SCSN removes the within-stage signature

Applied to the same checkpoints, SCSN reduces the within-stage probe to 0.465 on Sleep-EDF EEGNet and 0.209 on HMC EEGNet, differences of -0.273 [-0.299, -0.247] and -0.318 [-0.342, -0.294], $p < 10^{-4}$ and $p = 0.0020$; the reduction is present in all fifteen and all ten paired runs. The overall probe falls with it, to 0.314 and 0.073, but the within-stage number is the informative one. Figure 1 shows that the reduction holds in every stage: on Sleep-EDF the SCSN probe lies between 0.38 and 0.56 across wake, N1, N2, N3 and REM, against 0.65–0.82 for the uncorrected model; on HMC it lies between 0.15 and 0.32, against 0.45–0.59. The reduction is therefore not a consequence of a change in the predicted stage distribution.

The accuracy result follows the same direction but is smaller. On Sleep-EDF EEGNet SCSN changes accuracy by $+0.002$ [-0.004, $+0.009$], $p = 0.36$, a difference indistinguishable from zero; on HMC it gains $+0.019$ [$+0.010$, $+0.028$], $p = 0.0020$, over the uncorrected model, whereas the marginal correction *loses* 0.023 [-0.042, -0.005], $p = 0.010$. The HMC pattern is the one the algebra predicts: marginal adaptation removes the offset $a_{\mathcal{R}}$ but adds the stage-mix residual $\mu_{\text{coh}} - \bar{\mu}_{\mathcal{R}}$, which perturbs the logit margin, while SCSN maps the conditional means back to the prototypes and restores the margin. The Sleep-EDF cohort is smaller and more homogeneous, and the accuracy

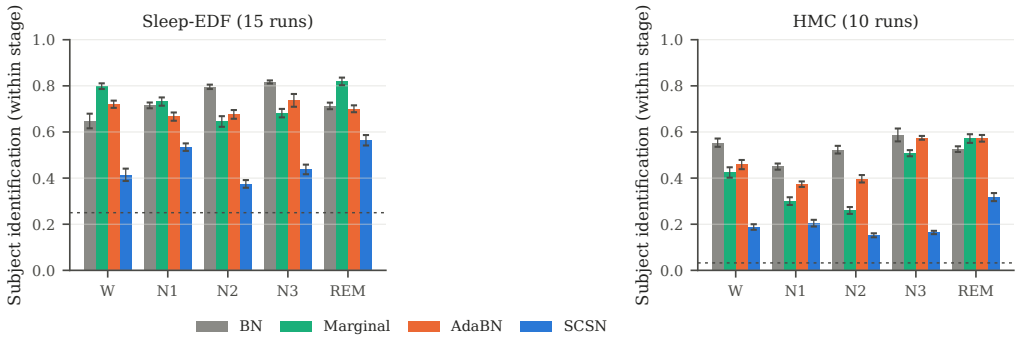


Figure 1: Subject identification within each stage, mean over runs (Sleep-EDF: fifteen runs; HMC: ten runs), for the uncorrected model (BN), the post-hoc marginal correction, layerwise AdaBN and SCSN. Error bars are standard errors over runs; the dashed line is chance. In two HMC runs a held-out recording contributes fewer than five N3 epochs, so the N3 probe is undefined there; the HMC N3 mean is over eight runs and the other HMC stages over ten. The marginal correction moves the within-stage probe little on Sleep-EDF and partway on HMC; SCSN reduces it in every stage.

effect there is not detectable.

5.3. Layerwise AdaBN: what it removes and what it leaves

Layerwise AdaBN is the stronger marginal baseline: it re-estimates the statistics of every normalization layer on the recording being scored and is nonlinear. Under the matched probe protocol it changes the overall probe from 0.715 to 0.466 and the within-stage probe from 0.738 to 0.700 on Sleep-EDF EEGNet, and from 0.385 to 0.212 and 0.527 to 0.473 on HMC EEGNet. It therefore removes part of the conditional signature as a side effect, but less than SCSN: the paired difference in within-stage probe between SCSN and AdaBN is -0.235 $[-0.266, -0.205]$ on Sleep-EDF ($p < 10^{-4}$) and -0.264 $[-0.278, -0.250]$ on HMC ($p = 0.0020$). On HMC it also removes less than the post-hoc marginal correction, which moves the probe from 0.527 to 0.411; layerwise statistics replacement is not a monotone improvement on a single-vector correction.

Figure 2 shows the per-run paired structure behind these averages: every run moves down from the uncorrected model to SCSN, and the AdaBN points sit between the two.

The residual after AdaBN is informative about its nature. Applying SCSN to the features produced by layerwise AdaBN changes the within-

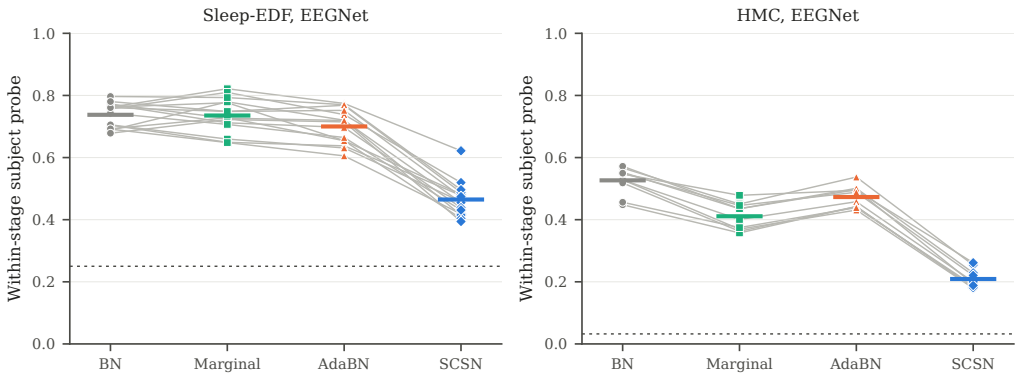


Figure 2: Within-stage subject probe for every run (grey lines), with the arm mean marked. Left: Sleep-EDF EEGNet, fifteen runs; right: HMC EEGNet, ten runs. The uncorrected model, the marginal correction, layerwise AdaBN and SCSN share the checkpoint and the probe subsample of each run.

stage probe by -0.0004 $[-0.0016, +0.0009]$ ($p = 0.85$), a difference that is not distinguishable from zero: once every layer has been adapted to the recording, what remains is not a stage-conditional mean shift, and the correction that targets a mean shift has nothing to remove. The two methods are therefore alternatives that remove different components, not two stages of one pipeline. This is consistent with the layerwise operation: re-estimating the statistics of a batch-normalized layer rescales and recentres its output per recording, so any additively stage-conditional component in that output is partly absorbed; what survives is carried by the higher-order structure of the representation, which a first-moment correction cannot reach.

5.4. Accuracy, and where the methods separate

SCSN is not the most accurate arm on every setting, and we report the ordering rather than selecting the setting that favors it. On Sleep-EDF EEGNet, the uncorrected model, the marginal correction, AdaBN and SCSN all lie within 0.007 of one another in accuracy, and the paired differences are not significant (Table 3). On HMC EEGNet, SCSN is the most accurate arm, 1.9 points above the uncorrected model and 4.2 points above the marginal correction. On Sleep-EDF SleepCNN, layerwise AdaBN is more accurate than SCSN by 0.014 $[-0.033, +0.060]$, a difference that is not significant ($p = 0.81$), and the uncorrected model is not significantly different from SCSN. On HMC SleepCNN the three arms are closer: SCSN reaches 0.712

against 0.698 for the uncorrected model and 0.702 for AdaBN, a significant gain over the former (+0.014 [+0.006, +0.022], $p = 0.004$) and a margin over the latter that is not significant (+0.010 [−0.001, +0.021], $p = 0.084$). The identity ordering is the one that is stable across settings: SCSN has the lowest within-stage probe in every one of the four batch-normalized settings, by margins between 0.27 and 0.44 over the uncorrected model and between 0.24 and 0.35 over layerwise AdaBN.

The Tent baseline is competitive on accuracy and does not remove the stage-conditional identity. On Sleep-EDF EEGNet, SCSN reaches an accuracy of 0.784 against 0.782 for Tent with the published optimizer on the same fifteen runs, a difference of +0.001 [−0.006, +0.008], $p = 0.85$, while its within-stage probe is 0.465 against 0.714 for Tent, a difference of −0.249 [−0.279, −0.220], $p < 10^{-4}$. On HMC EEGNet SCSN is 1.1 points more accurate (+0.011 [+0.001, +0.021], $p = 0.037$) with a within-stage probe of 0.209 against 0.453. On Sleep-EDF SleepCNN Tent is 1.8 points more accurate, a difference that is not significant ($p = 0.44$), but its within-stage probe is 0.858 against 0.550. Tent therefore keeps the patient signature in every setting in which it was run. Tent optimizes a surrogate on the affine parameters of the normalization layers; it is a batch-normalization method, and adapting it requires a separate design for group-normalized checkpoints. The comparison is therefore reported for the batch-normalized setting, and the group-normalized experiment isolates the applicability of the proposed correction.

5.5. Class agreement, calibration and per-stage behavior

An accuracy gain could in principle come from predicting the most frequent stage more often. The chance-corrected and class-balanced measures in Tables 1 and 2 rule that out: Cohen's κ and the macro-averaged F_1 follow accuracy in every arm. Where SCSN raises accuracy it raises both, by 0.023 [+0.012, +0.035] in κ and 0.018 [+0.008, +0.028] in macro-F_1 on HMC EEGNet (paired $p = 0.002$ and $p = 0.010$), and by 0.017 [+0.006, +0.027] and 0.013 [+0.005, +0.020] on HMC SleepCNN ($p = 0.014$ and $p = 0.006$). On Sleep-EDF the differences are within noise: κ changes by +0.003 ($p = 0.36$) on EEGNet and by −0.005 ($p = 0.62$) on SleepCNN. The HMC gain is therefore agreement across classes, not a reallocation toward the most frequent stage.

The per-stage recalls show where the accuracy difference comes from. On HMC, the marginal correction buys N2 recall at the expense of wake and

REM: on EEGNet it raises N2 recall by 0.079 ($p = 0.002$) while lowering wake by 0.107 ($p = 0.002$) and REM by 0.135 ($p = 0.014$), and on SleepCNN the same pattern is +0.063 for N2 ($p = 0.002$) against −0.174 for wake ($p = 0.002$) and −0.036 for N1 ($p = 0.037$), with the losses in N3 and REM in the same direction but not significant. The residual added by a single-vector correction is the same for every epoch of the night, so it cannot be stage-specific; on these checkpoints it shifts the decision toward N2, the most frequent stage. SCSN, which removes the stage-conditional shift, raises N2 recall by 0.040 ($p = 0.006$) on EEGNet and by 0.028 ($p = 0.006$) on SleepCNN without a significant loss in any other stage. Its gain is an improvement of the decision boundary the head was trained on, not a trade of one stage for another.

Calibration moves against accuracy on the small cohort. On Sleep-EDF EEGNet SCSN raises the expected calibration error from 0.030 to 0.044 (+0.013 [+0.006, +0.021], $p = 0.002$). The increase is not specific to SCSN — layerwise AdaBN reaches 0.043 on the same runs — but it is a cost of correcting the representation, and it is the reason the limitation in Section 6 recommends recalibration. On HMC the change is +0.006 ($p = 0.049$) on EEGNet and +0.002 ($p = 0.70$) on SleepCNN, and on the group-normalized HMC EEGNet SCSN lowers the error by 0.009 ($p = 0.002$). Conformal coverage is unaffected in every setting because the quantile is refitted on the calibration split after the correction.

5.6. Controls

Table 4 reports the controls on the two EEGNet settings. The interpolated family shows that identity removal is monotone in λ: on Sleep-EDF the within-stage probe falls from 0.673 at $\lambda = 0.25$ to 0.465 at $\lambda = 1$, while accuracy moves within 0.004; on HMC it falls from 0.434 to 0.209, while accuracy moves within 0.010 (Figure 3). The correction therefore buys identity without a measurable accuracy trade-off on either cohort, and the endpoint $\lambda = 0$ is the marginal correction whose within-stage probe is unchanged.

The rank restriction separates the two effects. Restricting the correction to the top principal directions of the between-subject, within-stage covariance removes much less identity than the full correction: $r = 4$ leaves a within-stage probe of 0.647 on Sleep-EDF and 0.347 on HMC, against 0.465 and 0.209 for the full correction, and $r = 1$ leaves 0.726 and 0.481. Low rank does not buy accuracy: the full correction has the same accuracy as $r = 4$ on both cohorts ($p = 0.93$ and $p = 1.0$). The conditional signature is therefore

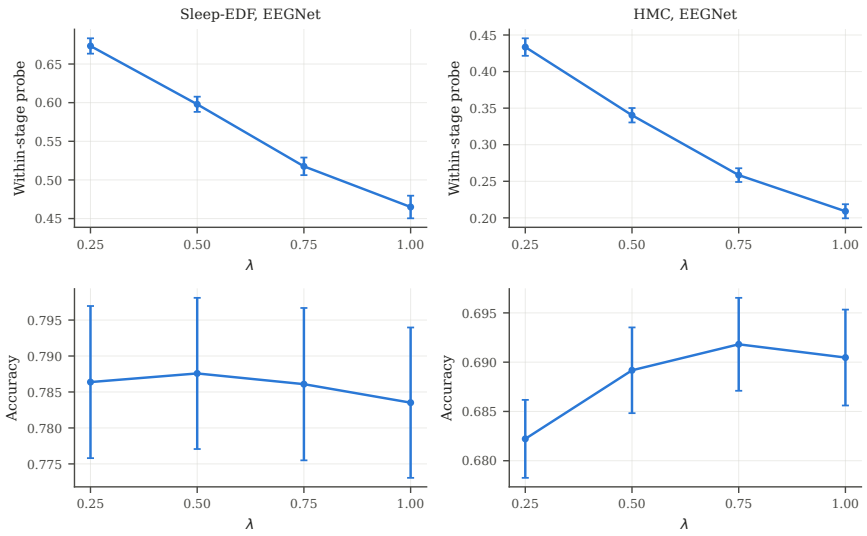


Figure 3: The interpolated family: within-stage probe (top) and accuracy (bottom) as a function of λ, for Sleep-EDF (left) and HMC (right). $\lambda = 1$ is SCSN, $\lambda \to 0$ is the marginal correction. Identity removal is monotone; accuracy is flat.

not confined to a few directions, and a practitioner should not restrict the correction for efficiency.

The orthogonalised control projects each estimated shift off the row space of the classifier, so the logits are unchanged exactly. Its accuracy is identical to the uncorrected model on both cohorts, as it must be, and its within-stage probe is *higher* than the uncorrected model's (0.792 against 0.738 on Sleep-EDF, 0.559 against 0.527 on HMC). The projection removes the component of the shift that the classifier reads and leaves the component it ignores, which is pure subject information; the probe sees more of it. This is the clearest evidence that the identity change in the SCSN arm is not a by-product of a change in predictions.

Two further controls bound the mechanism. Hard pseudo-labels over-correct: they reduce the within-stage probe slightly more than the soft posterior (0.402 against 0.465 on Sleep-EDF, 0.171 against 0.209 on HMC) but cost 1.3 and 2.2 points of accuracy. Mismatched statistics — applying the shifts of another recording of the same fold — remove less identity (0.568 and 0.718 within-stage) and cost accuracy (-0.022 and -0.059 relative to SCSN, both significant). The effect therefore requires the recording's own stage-conditional statistics, not merely a perturbation of the representation.

Table 4: Controls on the batch-normalized EEGNet settings: mean accuracy, within-stage probe, worst-patient conformal coverage and coverage dispersion over fifteen runs (Sleep-EDF) and ten runs (HMC).

	Sleep-EDF, EEGNet				HMC, EEGNet			
Arm	Acc	Within	Worst	SD	Acc	Within	Worst	SD
BN	0.781	0.738	0.828	0.055	0.672	0.527	0.649	0.086
Marginal	0.777	0.735	0.838	0.047	0.648	0.411	0.599	0.096
SCSN-$\lambda_{0.25}$	0.786	0.673	0.837	0.049	0.682	0.434	0.665	0.079
SCSN-$\lambda_{0.5}$	0.788	0.598	0.845	0.043	0.689	0.340	0.680	0.072
SCSN-$\lambda_{0.75}$	0.786	0.518	0.851	0.039	0.692	0.258	0.678	0.069
SCSN ($\lambda = 1$)	0.784	0.465	0.857	0.036	0.690	0.209	0.678	0.067
SCSN-r_1	0.783	0.726	0.854	0.039	0.676	0.481	0.686	0.072
SCSN-r_2	0.783	0.695	0.857	0.033	0.685	0.435	0.678	0.069
SCSN-r_4	0.784	0.647	0.856	0.036	0.690	0.347	0.682	0.067
SCSN-orth	0.781	0.792	0.828	0.055	0.672	0.559	0.649	0.086
SCSN-hard	0.770	0.402	0.863	0.032	0.669	0.171	0.668	0.072
SCSN-mismatch	0.761	0.568	0.806	0.066	0.632	0.718	0.618	0.095

The shrinkage constant λ_s is the only hyperparameter of the correction, and it is fixed at 10 throughout. Re-running the SCSN arm with $\lambda_s \in \{1, 100\}$ on the same checkpoints and the same cached representations, so that only the pseudo-count changes, shows where the value matters. Between $\lambda_s = 1$ and 10 the arm moves by at most 0.001 in accuracy and 0.006 in the within-stage probe on all four batch-normalized settings. At $\lambda_s = 100$ accuracy is again within 0.004 points, but the within-stage probe rises by 0.06 to 0.11: the pseudo-count now pulls the estimated per-stage means towards the cohort prototypes, and the correction removes less of the signature. The value used is therefore not tuned to the reported result. Any pseudo-count between one and ten gives the same arm, and the failure mode at a hundred is over-shrinkage, not sensitivity.

5.7. Group-normalized stagers

The group-normalized checkpoints are the setting in which standard AdaBN cannot be applied because there is no test-time batch statistic. A bespoke entropy objective on group-normalization affine parameters is a different method and is not evaluated here. The post-hoc marginal correction applies, and it removes the overall probe, but it moves the within-stage probe by between a tenth and a third of what SCSN removes, and not consistently in

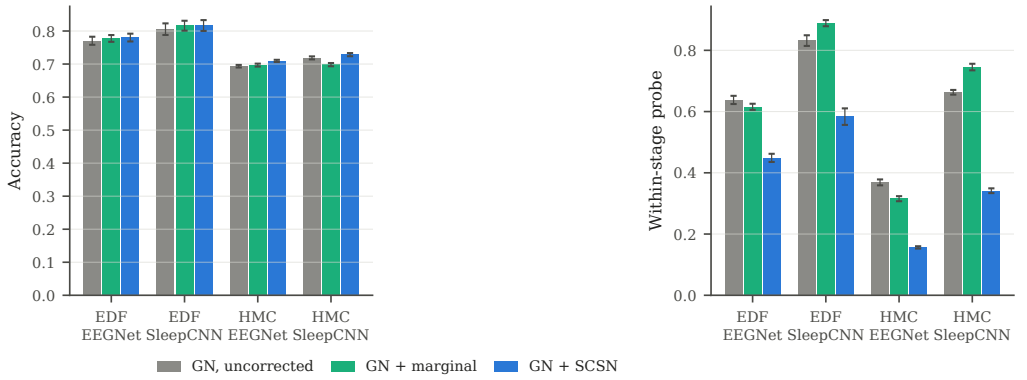


Figure 4: Group-normalized checkpoints: accuracy (left) and within-stage subject probe (right) for the uncorrected model, the post-hoc marginal correction and SCSN. Error bars are standard errors over runs. The marginal correction moves accuracy and the overall probe but barely moves the within-stage probe; SCSN reduces it.

sign: on Sleep-EDF EEGNet it changes the within-stage probe from 0.638 to 0.616 (paired $p = 0.030$) and on Sleep-EDF SleepCNN from 0.832 to 0.889, against 0.449 and 0.583 under SCSN (paired changes of -0.189 [-0.205, -0.174] and -0.248 [-0.284, -0.213]). On HMC EEGNet the uncorrected, marginal and SCSN within-stage probes are 0.369, 0.315 and 0.156, and on HMC SleepCNN they are 0.663, 0.745 and 0.341; on the latter the marginal correction moves the probe in the wrong direction (Figure 4).

The accuracy effect points the same way and is smaller than on the batch-normalized HMC model: SCSN gains $+0.010$ [$+0.005$, $+0.015$] ($p = 0.003$) over the uncorrected group-normalized EEGNet and $+0.011$ [$+0.001$, $+0.022$] ($p = 0.13$) over SleepCNN on Sleep-EDF, and $+0.016$ [$+0.013$, $+0.019$] and $+0.010$ [$+0.006$, $+0.014$] (both $p = 0.0020$) over HMC EEGNet and SleepCNN; on HMC SleepCNN the marginal correction *loses* two points. SCSN improves the worst-covered patient on the Sleep-EDF checkpoints (0.855 to 0.864 on EEGNet and 0.860 to 0.870 on SleepCNN), although the marginal correction improves it there too. On HMC EEGNet it improves the worst-covered patient from 0.695 to 0.708 and the dispersion from 0.067 to 0.059, and on HMC SleepCNN from 0.677 to 0.697 and from 0.070 to 0.059, although on both the marginal correction improves the worst patient further (0.724).

The result is the one the normalization argument predicts. Group normal-

ization removes the batch from the computation, so the representation of an epoch depends only on that epoch; it does not follow that the representation is free of the recording. The network that follows the normalized layers maps the remaining recording-dependent variation into the penultimate space, and a linear probe recovers the patient from it. Among the statistics-replacement approaches compared here, SCSN is the only stage-aware correction; it works without a batch statistic, optimizer, or weight update.

5.8. Patient-level conformal reliability

At $\alpha = 0.1$ every arm covers at close to the nominal rate on average, and the average hides the patient dependence. On Sleep-EDF EEGNet the uncorrected model has a per-patient coverage standard deviation of 0.055 and a worst-covered patient of 0.828; the marginal correction improves them to 0.047 and 0.838, and SCSN to 0.036 and 0.857. On HMC EEGNet the same ordering holds against the uncorrected model: SCSN improves the dispersion from 0.086 to 0.067 and the worst patient from 0.649 to 0.678, while the marginal correction *worsens* both (0.096 and 0.599). Layerwise AdaBN also improves the worst-covered patient over SCSN in both settings (0.866 against 0.857 on Sleep-EDF; 0.698 against 0.678 on HMC), and its dispersion is smaller on Sleep-EDF (0.028 against 0.036) and comparable on HMC (0.069 against 0.067). We report that rather than the comparison that favors the method: on a batch-normalized model, re-estimating every layer is a strong reliability intervention. On the group-normalized checkpoints, where standard AdaBN is unavailable, SCSN improves both quantities over the uncorrected model (Table 1).

Figure 5 shows the empirical distribution of per-patient coverage. Below the nominal level the uncorrected curve carries more mass than the SCSN curve, and the marginal curve is the worst of the three on HMC; above the nominal level the SCSN curve is the steeper one, so the shortfall of the patients that do fall short is smaller. The review workload moves with the sets: on Sleep-EDF EEGNet the fraction of multi-stage prediction sets falls from 0.750 to 0.729 under SCSN and to 0.741 under the marginal correction.

5.9. Transfer to a second centre, and the precondition of the method

A checkpoint trained on Sleep-EDF and never shown an HMC epoch is evaluated on all 137,243 HMC epochs under the source statistics, the post-hoc marginal correction, SCSN, layerwise AdaBN and SCSN stacked on Ad-

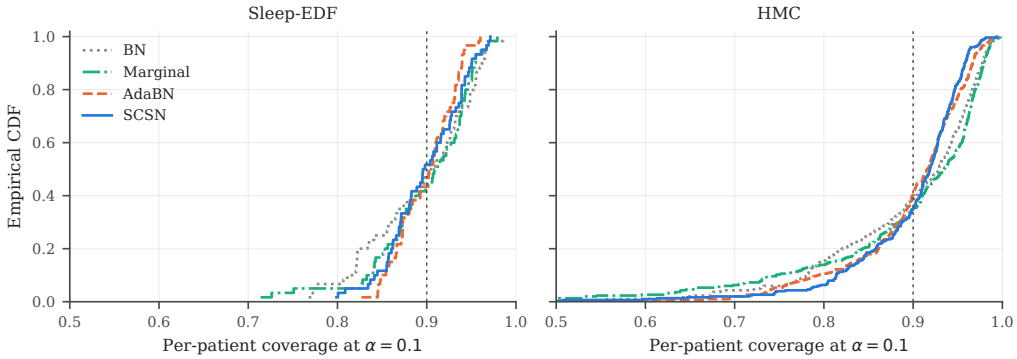


Figure 5: Empirical distribution of per-patient conformal coverage at $\alpha = 0.1$, pooled over runs. The dashed line is the nominal level. The uncorrected model has more mass below the nominal level; the marginal correction moves the distribution in the wrong direction on HMC, and SCSN moves it in the right direction on both cohorts.

aBN (Table 5). This is a transfer across population, montage, equipment, sampling rate and scoring manual, not merely across patients.

Under the source statistics the model reaches an accuracy of 0.475 with a worst-recording accuracy of 0.128. The marginal correction raises accuracy to 0.505; layerwise AdaBN raises it to 0.553 and the worst recording to 0.214. SCSN reaches 0.489, below both. The oracle variant, which uses the true stages to estimate the shifts, reaches 0.727: the estimator is not the limitation, the posterior is. When the source model is applied to a cohort it was not trained on, its posterior is weak, the stage-conditional means estimated from it are dominated by the prior and the recording offset, and the correction behaves like the marginal one. SCSN should therefore not be applied blindly across cohorts; the precondition is a posterior informative enough to index the stages, and it can be checked on the unlabeled night before the correction is applied.

Two caveats bound this experiment. Every arm is evaluated on a single checkpoint, and no subject probe is computed on the transferred representation, so the section speaks to accuracy and per-recording spread, not to identity. It shows that the intervention travels with the model to an unseen centre only when the posterior does; the within-cohort experiments of Sections 5.2–5.3 are what show that the identity decomposition is the same there.

Table 5: Cross-cohort transfer: a Sleep-EDF checkpoint evaluated on all HMC epochs. "Worst" and "SD" are the minimum and standard deviation of per-recording accuracy. The group-normalized checkpoints have no statistics to replace and are not part of this table.

	EEGNet		
	Acc	Worst	SD
Cohort statistics	0.475	0.128	0.150
Marginal	0.505	0.115	0.143
SCSN	0.489	0.114	0.156
SCSN-$\lambda_{0.5}$	0.497	0.113	0.152
AdaBN	0.553	0.214	0.134
AdaBN + SCSN	0.555	0.204	0.139
SCSN (oracle labels)	0.727	0.245	0.120

6. Discussion

6.1. What the decomposition buys

The literature on subject variability in physiological signal classifiers is mostly a literature about remedies: add a subject identifier, adversarially remove one, fine-tune on the new patient. Each remedy is expensive in a different currency — annotations, training time, or a labeled target cohort. The decomposition of Section 3 separates the patient into a recording-wide offset and a stage-conditional signature, and the experiments show that the two parts have different remedies. The offset is manufactured by the normalization layer and can be removed by replacing its statistics, with no optimizer and no label. The signature can be targeted only by an operation that knows the stage, and the inability of any marginal method to target it is visible in the within-stage probe.

The decomposition also changes what a subject-invariance claim should report. An overall probe conflates the two components: a single vector subtraction per recording drives it close to chance even when the conditional sig-

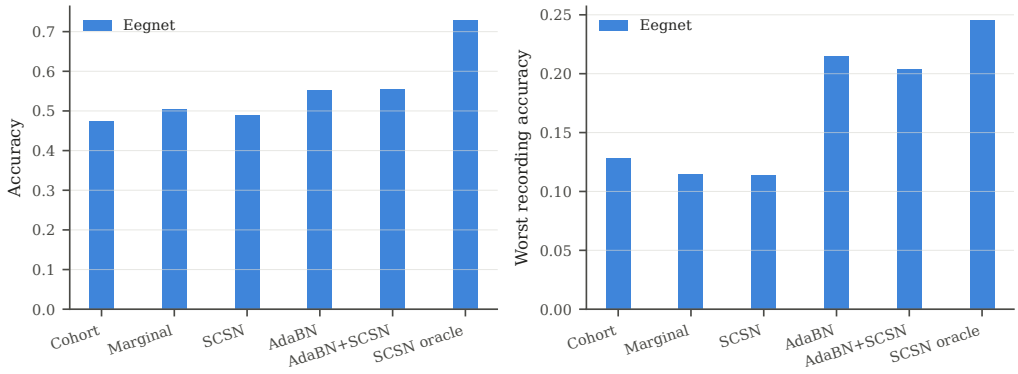


Figure 6: Cross-cohort accuracy (left) and worst-recording accuracy (right) for the transferred checkpoint. The oracle uses the true stages; the gap between it and SCSN is the cost of the weak target posterior.

nature is untouched, because the recording mean is the dominant direction. A paper that reports only the overall probe would conclude that marginal adaptation and SCSN are equivalent on HMC — both reach a probe near 0.07 — when the within-stage probe is 0.41 under the former and 0.21 under the latter. The within-stage probe is the quantity that matches the object the classifier uses: the relative position of two stage-conditional means.

6.2. Why marginal adaptation is not enough

Proposition 1 gives the residual of any correction that subtracts one vector per recording: $\mu_k + s_{\mathcal{R},k} + (\mu_{\text{coh}} - \bar{\mu}_{\mathcal{R}})$. The first two terms say that the class-conditional means are still those of the patient, not those of the cohort; the third says that the correction adds an offset determined by the patient's stage mix. The second term is why the within-stage probe survives. The third is why marginal adaptation can *lose* accuracy: on HMC, where recordings differ in stage mix and the cohort is large, the post-hoc marginal arm is 2.3 points worse than the uncorrected model. The loss is not spread evenly over the night: on HMC EEGNet the marginal arm gives up 0.135 of REM recall and 0.107 of wake recall for 0.079 of N2 recall, whereas SCSN raises N2 recall by 0.040 without a significant loss in any stage (Section 5.5). An oracle marginal correction that knew the recording's stage mix would remove the third term but not the second; the signature is out of reach of any operation that uses one mean per recording.

SCSN removes both terms with the model's own posterior. The price is that the posterior must be informative: the residual of the soft correction is the part of the signature assigned to the wrong stage, bounded by the posterior error. The cross-cohort experiment is the regime in which that price is not paid, and the degradation there is the clearest evidence that the mechanism is the one the algebra describes rather than an unspecified regularisation effect.

6.3. What kind of claim this is

The method can be read as an inference-time trick, and that reading would miss what is being claimed. Nothing here changes the training procedure, the architecture or the objective; the operation is arithmetic on a frozen representation, and it is not presented as a new architecture or a new learning objective. The claim is nevertheless about the model: that the stage-conditional means of its penultimate representation are shifted by the patient, that the shift is identifiable from the recording and the model's own posterior, and that no operation using a single vector per recording can remove it (Corollary 1). That is a statement about what the network has learned, in the same sense as a probing result, and the paper makes it falsifiable with a measurement it introduces: the within-stage probe falls under SCSN and does not fall under any marginal correction, in every setting tested. The estimator then follows from the statement through the conditional expectation of the shift given the model's posterior (Remark 1), and the closed form follows from the deployment setting in which such a statement is useful: a checkpoint whose weights are frozen and a night whose labels are unknown. A method that needed a second pass, an optimizer or a gradient through the posterior would answer the same analytical question and be a different engineering object. The object of the contribution is the analysis and the measurement. The algorithm is what that analysis leaves as the only available operation; its deployment properties are what make the analysis usable at the bedside, and they are not where the novelty lies.

6.4. When the correction is free and when it is not

On Sleep-EDF EEGNet, SCSN leaves accuracy where the uncorrected model and layerwise AdaBN leave it; on HMC EEGNet and HMC SleepCNN it gains 1.9 and 1.4 points over the uncorrected model and has the highest mean accuracy of the arms, although on HMC SleepCNN the margin over layerwise AdaBN is not significant ($p = 0.084$). The HMC gains survive the

chance-corrected and class-balanced measures: κ rises by 0.023 and 0.017 and macro-F_1 by 0.018 and 0.013 on the two backbones, with no comparable change on Sleep-EDF. On Sleep-EDF SleepCNN, layerwise AdaBN is 1.4 points more accurate than SCSN, a difference that is not significant. We report the ordering rather than selecting the cohort on which the method wins. Two mechanisms plausibly explain the ordering. Layerwise AdaBN is nonlinear and acts at every layer: it removes part of the conditional signature as a side effect, lowering the within-stage probe by 0.04 to 0.09 where it acts at all, and not at all on Sleep-EDF SleepCNN, but it has no stage index and cannot target that signature; SCSN removes 0.24 to 0.35 more. And the accuracy effect of SCSN depends on the posterior: where the posterior is weak, the correction approaches the marginal one, as the cross-cohort experiment shows. Neither mechanism makes SCSN preferable on accuracy alone on Sleep-EDF; the case for SCSN there rests on the identity it removes and on the fact that the same operation is the only one available on a group-normalized checkpoint.

Tent is a strong baseline on Sleep-EDF and we do not claim to beat it on accuracy. It is also a batch-normalization method: it optimizes the affine parameters of batch-normalization layers. Applying it unchanged to a group-normalized model is not defined, and a GroupNorm-specific entropy method would require a separate design and tuning protocol. The comparison is therefore reported for the batch-normalized checkpoints where both methods are defined.

6.5. The group-normalized setting

Group normalization is often described as removing subject information because it removes the batch. The experiments show that it does not remove the patient: a linear probe identifies the patient from a group-normalized representation well above chance on both cohorts and both backbones. What it removes is the batch dependence of the normalization, not the recording-dependent variation that the following layers map into the penultimate space.

This is the setting in which the contribution is least ambiguous within the family considered here. Standard AdaBN is undefined because there is no test-time batch statistic, and standard Tent targets batch-normalization affine parameters. The post-hoc marginal correction is defined and removes the overall probe, but it moves the within-stage probe by between a tenth and a third of what SCSN removes, and in the wrong direction on both SleepCNN checkpoints, because a single vector per recording carries no stage index.

SCSN is the only stage-aware statistics-replacement correction among those evaluated, and on the group-normalized checkpoints it improves accuracy and the worst-covered patient over the uncorrected model. A practitioner who chooses group normalization for its batch independence therefore has a label-free inference-time correction available without changing the architecture.

6.6. Reading the reliability result correctly

The conformal guarantee is marginal over patients, and the numbers make the sense in which it is marginal concrete: every arm covers at about the nominal rate on average, while the worst-covered patient of a fold is covered well below it. The dispersion of per-patient coverage is the quantity that separates a model whose guarantee is uniform from one whose guarantee is an average.

The correction that removes the patient signature also reduces the concentration of failures, but not uniformly across settings. On the group-normalized checkpoints SCSN improves the worst-covered patient and the dispersion in all four settings. On the batch-normalized EEGNet checkpoints, layerwise AdaBN attains a better worst-covered patient than SCSN (0.866 against 0.857 on Sleep-EDF and 0.698 against 0.678 on HMC) and a smaller dispersion on Sleep-EDF (0.028 against 0.036), with the two dispersions comparable on HMC (0.069 against 0.067); on HMC SleepCNN the ordering reverses, with SCSN better on both quantities (0.664 against 0.649 and 0.066 against 0.072). We report that rather than the comparison that favors the method. The safe statement is the mechanistic one: removing the component of the representation that carries the patient reduces the patient-dependence of the coverage, and the size of the reduction depends on how much of that component the correction removes.

6.7. The boundary of the method

The cross-cohort experiment defines the regime in which a stage-conditional correction should not be used. When a model trained on one centre is applied to another, its posterior is weak, and the stage-conditional means estimated from that posterior are dominated by the prior and by the recording offset. SCSN then performs like the marginal correction and below layerwise AdaBN; the oracle-label version, which uses the true stages to estimate the shifts, recovers a large part of the gap, so the limitation is the posterior rather than the estimator. This is a useful diagnostic: the method has a measurable precondition, and the precondition can be checked on the

unlabeled night before the correction is applied, for example by the entropy or the confidence of the uncorrected posterior.

6.8. Limitations

Cohort size. Sleep-EDF has 20 subjects, so a fold holds out four of them and the probe is a four-way problem. A four-way probe can only be so informative, and a subject-level statistic computed over four subjects is noisy. This is the reason the paper carries a second cohort of 151 subjects; the second cohort is larger than most in the literature but comes from one centre, one montage and one scoring team, and it is a cohort of patients referred for suspected sleep disorders rather than of healthy volunteers.

Probe capacity. The probe is a linear readout. A nonlinear probe would find more identity in every arm, and the fractions reported here are lower bounds on what the representation contains rather than statements about what it contains. The comparison between arms is not affected by this, since the probe is the same object in every arm, but the absolute level should be read as a floor.

Backbones. Two convolutional backbones are used on both cohorts, chosen to differ in depth, width and receptive field. Neither is a transformer, and the extent to which normalization-driven identity matters in attention-based stagers is not addressed here.

Posterior quality. The correction is only as good as the posterior that indexes the stages. On the within-cohort experiments the posterior is informative enough that the correction is close to exact; across cohorts it is not, and the method degrades to the marginal correction. A stager with a poorly calibrated head would need the head recalibrated before the correction is applied.

Probability calibration. Re-estimating the statistics, or correcting the representation, changes the scale of the logits relative to the scale on which the posterior head was fitted, and the expected calibration error moves with it: on Sleep-EDF EEGNet from 0.030 to 0.044, with layerwise AdaBN at 0.043 on the same runs, and by at most 0.006 on HMC. Conformal sets are unaffected in coverage because their quantile is refitted on the calibration split after the same intervention, but a user who reads the raw probabilities of a deployed stager should recalibrate them after the correction.

Offline operation. As with AdaBN, the statistics are estimated on the whole night. Clinical staging is scored after the night, so the constraint is not binding in the application we target, but a real-time monitor would have to estimate the statistics on a prefix, which we have not tested.

GroupNorm-specific test-time adaptation. Because group normalization has learnable affine parameters, an entropy-based test-time objective could be designed for it. We did not implement that baseline here: doing so would introduce a new optimizer, step-size schedule, and stopping rule, and would no longer be the standard Tent method. The group-normalized comparison therefore establishes applicability and internal mechanism, not superiority to every possible GroupNorm-specific adaptation objective.

Single channel. Both cohorts are staged from a single EEG derivation. Multi-channel stagers have more ways to encode a patient, and the balance between the marginal and the conditional part of the identity may differ.

Statistical inference. The Wilcoxon tests are exploratory and pairwise unadjusted. Seeds within a fold reuse the same held-out subjects, so paired runs are not fully independent; fold-level sign counts are therefore reported alongside the run-level tests to reduce reliance on pseudo-replication. We make no familywise-error claim. The consistency of effect directions across cohorts, backbones, and normalization schemes, rather than any single p-value, is the basis for the conclusions.

6.9. Practical recipe

For a stager that is already trained and validated, the procedure the results support is short.

1. Fit a linear probe from the penultimate representation to the patient, overall and within stage. If the overall probe is near chance, the marginal patient is not in the representation. If the within-stage probe is near chance, the conditional signature is not there either, and no stage-conditional correction is needed.
2. If the within-stage probe is high, apply SCSN on each recording before predicting it. On a batch-normalized model, compare with layer-wise AdaBN; on a group-normalized model, SCSN is the only stage-conditional option.

3. Check the uncorrected posterior before applying the correction. If it is close to uniform, the stage-conditional means are not identifiable and the correction should be reduced to the marginal one.
4. Recalibrate the conformal score on recordings processed the same way, and report per-patient coverage dispersion beside average coverage.

The first step is a diagnostic that costs one logistic regression; the second adds a few matrix products to the forward pass the stager performs anyway; the third is a scalar check on the posterior; the fourth is bookkeeping. None of them requires a labeled target cohort, and all of them can be applied to a model whose weights will never be touched again.

7. Conclusion

A stager's representation of a patient has two parts. One is a recording-wide offset manufactured by the normalization statistics; the other is a stage-conditional signature that describes how this patient's sleep stages differ from the cohort prototypes. Marginal adaptation removes the first and, as the within-stage probe shows, cannot target the second. Layerwise adaptation may absorb part of it as a nonlinear side effect, but a stage-conditional operation is required to remove the signature explicitly. Stage-conditional shift normalization removes it with the model's own posterior, in closed form, at the cost of one pass over the unlabeled night.

The experiments support three statements. The within-stage probe separates the two components where the overall probe cannot, and it is the measurement a subject-invariance claim should report. SCSN reduces that probe in every setting, by between 0.19 and 0.44, and on the group-normalized checkpoints it is the only stage-aware statistics-replacement method evaluated that does so. Where it changes accuracy, it improves it: significantly on the HMC backbones and on three of four group-normalized settings, not measurably on Sleep-EDF EEGNet, and within noise of layerwise AdaBN on both SleepCNN settings; the gains survive chance correction and class balancing, and they come with a better worst-covered patient and a smaller coverage dispersion.

The limitations define the next experiments. The method is evaluated on two single-channel convolutional backbones, its estimate is only as good as the posterior that indexes the stages, and it assumes the whole night is available. A multi-channel or attention-based stager, a recalibrated posterior,

and a prefix-based online variant are the natural places to test whether the decomposition — rather than the particular estimator — is what carries the result.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Sleep-EDF and the Haaglanden Medisch Centrum Sleep Staging Database are publicly available from PhysioNet at `https://physionet.org/content/sleep-edfx/1.0.0/` and `https://physionet.org/content/hmc-sleep-staging/1.1.0/`, respectively.

Code availability

The code that produces every number and figure of this paper, together with the trained-checkpoint evaluation scripts and released result summaries, is available at `https://github.com/YuhaoPeng1103/SleepSCSN`.

Ethics statement

This study used publicly available, de-identified Sleep-EDF and HMC recordings from PhysioNet. No new human-subject data were collected, and additional ethics approval was therefore not required.

CRediT authorship contribution statement

Zhiwen Song: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. **Yuhao Peng:** Supervision, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, manuscript organization, and consistency checks. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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